

Some Popular PNP Machines Available in the Market

Model	Vision system	Largest board size	Feeders	Applicable components	Speed (components per hour)
VisionBot	Fully-integrated computer vision software	Minimum 200mm x 150mm	24	Tested with 0402	2000
NeoDen-4	Intelligent flying vision	290mm x 1200mm	48	Minimum 0201; Maximum 30x30mm IC, BGA, QFN	12000 with vision off, 5000 with vision on
LPKF ProtoPlace S	Micro-camera	297mm x 420mm	12	0201 chip components	N/A
DDM Novastar L-SF40	Colour CCD card camera	343mm x 560mm	N/A	0201 - SOIC - PLCC down to 20mil pitch	1800
Autotronic BS281	Vision detection	150mm x 100mm	64	0201 devices, CSPs, μ BGAs, flip-chips, ultra-fine pitch 0.5mm for QFPs and even odd-form components	3000

work with a local distributor to get one.

We recommend that you go for it if you can afford it. So, what does it do?

1. Verifies that the part is picked up and is on the nozzle. It is easy for the machine to not pick up the part and keep going without picking up a part.

2. Aligns the board. When you start a job and put the board in the machine vision, it will look for marks on the board called fiducial markers. This becomes even more important as the parts become smaller.

3. Helps align the pickup points and identify where the feeders are. Even if you have the mechanical position of the parts correct, you could be off by a millimetre. The problem occurs when the part is just half a millimetre.

Neoden-4 has an intelligent vision system that allows edge detection of every component. It will analyse, add correction, co-ordinate and place accurately using micron high-resolution cameras for analysis and calculation.

Speed of placement per minute.

This is important as it decides the productivity that you could get out of your machine. Usually, low-end

machines are not going to do more than 10 or 20 pieces per minute. This should not be a problem unless you have a lot of placements to do. For systems with machine vision, switching it off should give you significant increase in the speed of placement since it now does not have to verify the placement of every component.

Other features that can be considered include the smallest size of the part that can be picked. Currently, one of the smallest is 0402 imperial, which translates to 1mm x 0.5mm.

Also important is the number of heads that the machine has. Venu Narayan Vaishnav, owner, Chipmax Group, says, "NeoDen-4 can achieve accuracy with 0201 chips and it is a quad-head machine that can carry four component pick-up nozzles at a time."

Another good example is VisionBot PNP. An interesting machine made by a maker for other makers, it is a comparatively inexpensive tool for PCB assembly that works with gerber and csv files. You can go for 24 to 96 feeders depending on the model, and it is being launched via *Kickstarter*.

Another interesting one is the PNP machine by ProtoVoltaics, which

can speed up the system by using compute unified device architecture (CUDA) accelerated computer vision from an NVIDIA graphics card to speed up the processing of video tasks like rotational vision correction.

The PCBGRIP electronics assembly system being offered is not a complete PNP machine, but holds PCBs, components, stencils and other items so that you can improve the quality and speed of your soldering.

Of stencils and printers

The paste you use along with a stencil is like toothpaste, but it is grey and has to be kept refrigerated in order to last at least six months. If it dries even after being kept in the fridge, you can add alcohol and rejuvenate it. You do not need a lot of it, though. To apply the paste, you need to have a stencil, a piece of material with holes.

Vaishnav adds, "Solder paste comes in different compositions, leaded and without lead to comply with the restriction of hazardous substances (ROHS) directive. There are a few good Chinese brands that are there in the market."

How do you build stencils? You can laser-cut these stencils pretty easily. Materials can be high-durability or high-temperature plastic. You can also use vellum, which is a translucent paper that you can get at a stationary store. It can be used for about 50 boards and can be easily laser-cut.

How do you select a laser cutter? Laser cutters have a limitation of the smallest hole these can cut. So this

How to DIY without a stencil printer

I use frameless stencils and make use of the same method I used when I was a hobbyist. Make a registration on the stencil and then use some waste PCB boards to make a mark. Then take a stencil and tape it onto the board like a hinge and line it up with tape. This way, when it comes down, it comes down at the same part of the board. So you do not really need to do anything so sophisticated; you just need a work surface that you can move around the place.

—Bob Coggeshall